A Review on Student Modeling Approaches in ITS

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***Abstract* – The Intelligent Tutoring System (ITS) is a software system used in the field of education to provide personalized and intelligent tutoring to the students based on their preferences, knowledge, purpose and profile. Based on a general consensus amongst researchers, an ITS consists of four components-domain model, student model, tutoring model and user interface model. Of the four components, student modeling is considered to be the building block in the field of ITS research because it can model the behavior of a tutor by making them understand how students learn. The challenge is to decide the most appropriate sequence of activities that can improve the average learning rate over all the skills. This paper analyses various knowledge tracing methods and empirical methods for optimized prediction of a student's mastery of a skill.**

Keywords – Bayesian Knowledge Tracing (BKT); Brute-Force Model; Expectation Maximization; Knowledge Components (KC); Performance factor Analysis (PFA); Intelligent Tutoring Systems (ITS).

I. INTRODUCTION

Intelligent Tutoring Systems (ITSs) are computer programs that are designed to provide an effective and adaptive learning and which also helps tutors to evaluate the thought processes of a learner to estimate what kind of help they need at that moment of time. A tutor must interpret and evaluate each student's action with respect to his prior actions. In addition to evaluating the student's action, an intelligent tutor should also suggest a next action to be taken, especially when a student is stuck while solving a problem. This suggestion may come in the form of a series of as-needed hints that get increasingly specific. To perform the evaluation and suggestion, the tutor uses an adaptive model that represents most probable solutions to the activity, which is used to summarize how a student's input is related to common misunderstandings, and provides the type of feedback or hints that helps the student the most, in completing the activity.

Interactive help could be provided to students, only if a tutoring system is aware of what skill set the student has and the type of skill the student is currently trying to achieve. These modeling tasks involve a high level of uncertainty as it is very difficult to analyze when and how a student's imagination and thought process varies. Most of the intelligent tutoring systems have been built through extensive knowledge engineering andevaluations.

The paper is organized as follows. A detailed review on various student modelling approaches is presented in Section II. In Section III, the paper describes Bayesian Knowledge Tracing methods and how they are used for tutoring. Section IV describes various challenges in estimating the Bayesian Knowledge Tracing parameters. The Section V discusses various approaches for BKT model and these approaches are evaluated in Section VI. This paper ends with conclusion and direction for future work.

II.REVIEW OF STUDENT MODELING APPROACHES

Many approaches have been followed to model student's knowledge and predict a student's skill set over the last few decades.The process of building a student cognitive model is very difficult because of the uncertainty that exists in the knowledge level of the students and their learning strategy. There are different approaches to build an effective student model, which vary from Overlay Models, to Bayes Nets, to Bayesian Knowledge Tracing [6], to models based on statistical models like Item-Response Theory such as Performance Factors Analysis (PFA). Let us see these approaches in brief.

A. Overlay Models

A domain model is divided into a set of knowledge entities and the overlay model simply records whether the learner has mastered the entity or not. If an entity A is dependent on entity B, we can deduce that the learner must understand A before learning B. This means that the learner model is the overlay with at most 1 for entity A. Overlay Model was considered to be very popular, with its simplicity in both concept and computation. Overlay models are easily built and managed. Although overlay factor helps tutor to infer that whether the skill has been mastered or not, it does not provide the way to infer user's knowledge from the evidence collected throughout the learning process of the user. But when overlay modeling is associated with other statistical approaches like Bayesian network, it can infer student's knowledge.

B. Bayesian Network

The Bayesian Network (BN) is a directed acyclic graph in which nodes signify variables and arcs signifies probabilistic dependence among the variables. The conditional probabilities

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of each node are considered as the parameters used to represent the uncertainty.

If we are using BNs to define a student model, the variables can represent different factors depending on the problem domain. The variables can be facts, rules, skills, concepts, problems, etc. These variables are linked by relationships between them, such as part-of, prerequisite-of, etc. The conditional probabilities must be specified, once variable and links are decided [11]. Highly accurate estimations of the student's cognitive state at all levels of learning process can be achieved through a Bayesian integrated model.

C. Correct First Attempt Rate (CFAR)

For the prediction of students mastery and knowledge level, the CFAR which stands for "Correct First Attempt Rate", can be used. Based on the student's average accuracy level on that skill till current moment, student performance on a given skill can be predicted. For example, CFAR can be calculated as:

CFAR = Number of attempts with studentid = sid and CFA = 1

CFA (Correctness on First Attempt) could be zero or one. For each step for a student, the above CFAR would be used to calculate a certain characteristic feature. This characteristic feature and CFA can be correlated, which is now the target for prediction.

D. Performance Factor Analysis(PFA)

PFA is a logistic regression model, which predicts accuracy based on the student's number of prior failures and prior successes on that skill. An overall difficulty parameter β is also fit for each skill [4].

The PFA equation is:

$$m(i, j \in KCs \in, s, f) = \beta_j + \sum(\gamma_j S_{i\,j} + \rho_j F_{i\,j}) \qquad (1)$$

Where m is a logit value which indicates a total learning rate for student i (ability captured by α parameter) using one or more Knowledge Components (KC) j. Logit models are used whenever the dependent variable is binary, which takes values 0 or 1. The difficulty level of a KC is determined by β parameter based on's' successes of student i with KC j. For each student, F is used to keep track of failures a student had for a particular Knowledge Component and γ and ρ are used to scale

E. Tabling

This approach can be used to check what percentage of students with that same pattern of performance gave correct answer to the next question from the prior responses. It is a very simple baseline approach that can be used for the prediction of student's performance [3]. In the training phase, a table is constructed for each skill: each row in that table represents a possible pattern of student performance in most recent data points from most recent dataset [2]. For example, if 3 questions are answered correctly, the row for that skillset would be 111. If the first two responses are wrong and the last response is correct then the row value would be 001. A value would then be calculated for each row corresponding to the skillset, based on which prediction can be made. Suppose 20 students have all the correct responses i.e., 111 and 3 students have the response 001, then the value for the row 001 could be (20/(20+3)) which is 0.94.

To predict a student's performance, research has been carried out by analyzing various student models. Whether it be a mere theoretical comparison or a statistical comparison, the results generated showed that there are considerable differences between each of the approaches. The more the accuracy of a model,the less practice a student requires in acquiring a skillset or a Knowledge Component [6].

A student might have some chance to do a question wrong, even though he knows how to do it. He also might have some chance of getting the question correct, even though he doesn't know how to do it. Based on these observations, a tutoring model can be built, where a tutor will not allow students to proceed without successfully demonstrating that they possess some desired skill (a KC) and once a student has acquired a skill, ideally questions pertaining to that skill will no longer be asked.A Bayesian network perfectly fits this model. Most of the ITSs uses Bayesian network for learning the students mastery over a skill because of its simplicity and accuracy. Further, Bayesian Knowledge Tracing approaches are analyzed in detail.

III. BAYESIAN KNOWLEDGE TRACING

Bayesian Knowledge Tracing (BKT) is a type of student model used in adaptive tutoring that infers a student's knowledge from previous responses of a student and helps to evaluate the future performance [5]. Typically, every skill has a BKT model. BKT assumes four probability factors, in order to calculate the student's mastery of a skill based on their performance history:

a. $P(S_0)$, the probability that a student is well versed in that skill, before attempting the first exercise..

b. $P(S_1)$, the probability that a student who is not aware or learned that skill, willprobably know it after the next exercise..

c. $P(S_2)$, the probability that a student who is not aware of the skill or learned that skill ,will answer a question correctly.

d. $P(S_3)$, the probability that a student will answer a question incorrectly despite knowing the skill.

According to this model, P(S) is probability that a student will be in the known state or might have learned a skill at time t+1 , if a student is in the unknown state or might not have learned that skill at time t, Additionally, BKT models typically assume that forgetting does not occur. Once a student is in the known state, they don't transit to unknown state.

The Bayesian Knowledge Tracing (BKT) model assumes that, for each skill, student is in either known state or in unknown state i.e., he might have either acquired that skill or might not

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have acquired the skill. Thus BKT uses a two-state learning model for each skill/knowledge component [5]. A student can jump from a known state to unknown stateduring each practice session, whether answers are right or wrong with a learning probability (S_1). If a student forgets the skill the probability is zero. It is based on the assumption that a student never transits from learned to unlearned stat. A student who has well mastered or learnt a skill might give a correct or a wrong answer (slip), which is represented by the probability (S_3). Similarly, a student who has no mastery over that skill can sometimes guessand make the correct response with probability (S_2). But if the student has mastery over that skill before even practice sessions, then that probability is represented by (S_0). After each practicing session, the system updates its estimate of student's knowledge state (S_n), based on the correct responses and the probability that a student have mastered a skill and it is as follows [3]:

$$P(S_{n-1})|Correct_n = \frac{P(S_{n-1})*(1-P(S_3))}{P(S_{n-1})*(1-P(S_3))+(1-P(P(S_{n-1})*(1-P(S_2)} \quad (2)$$

$$P(S_{n-1})|InCorrect_n = \frac{P(S_{n-1})*(P(S_3))}{P(S_{n-1})*(P(S_3))+(1-P(P(L_{n-1})*(1-P(S_2))} \quad (3)$$

$$P(S_n)|Action_n = P(S_{n-1})|Action_n + ((1 - \ (1 - P(S_{n-1}|Action_n) * P(S_1) \quad (4)$$

IV. CHALLENGES IN ESTIMATING PARAMETERS FOR BAYESIAN KNOWLEDGE TRACING MODELS

There are two issues that suffer from model fitting of Bayesian parameters. They are identifiability and degeneracy.

i.Identifiabilty: There could be different combinations of the above mentioned four Bayesian parameters that could be applied on same set of historical data, but it need not always give the same estimates. This issue is of significant importance because each of these parameters requires different amounts of practice sessions before reaching to the assumption that a student has masters that skill[11].

ii.Degeneracy: Knowledge tracing in Bayesian network is based on the concept that if a student knows the skill, he would correctly perform and if a student correctly performs, then he has acquired that skill. This assumption is irrelevant if a student performs correctly by mere guess. This may cause the model to degenerate. For instance, a model would degenerate, if a guess (G) parameter or slip (S) parameter is greater than 0.5. A student who has mastered a skill and who is more likely to get a wrong answer than a correct answer has always a slip parameter over 0.5. Similarly, a guess parameter over 0.5 implies that a student who hasn't mastered or learnt a skill is more likely to get a correct answer than a wrong answer.

To overcome these challenges, there are various approaches by which parameters are fit to the BKT, which is covered in next section.

V. MODEL FITTING APPROACHES FOR BKT MODEL

The major approaches by which parameters are fit to the BKT are performed by the baseline approach, bounded guess and slip approach and Dirichlet priors [1]. These approaches are detailed below.

i. The Baseline Approach: It takes any value between 0 and 1. The toolkit used for this approach is Bayes Toolkit-Student Modeling (BNT-SM) [12]. It includes specifying the relationships of factors that can gauge a student's knowledge in an XML specification file along with the computational data. Even though BNT-SM offers flexible and more plausible model parameters, there exist some restrictions on the possible modeling of relationships. For instance, BNT-SM approach is restricted to a specific time slice. It does not consider the factors that go backward in time or that skip forward past the next time slice. Currently, BNT-SM handles any unobserved and observed variables at the current time slice, but these variables are limited to discrete values [9].

ii. The Bounded Guess and Slip Approach: This approach binds the guess and slip parameter to a specific range and makes model degeneracy impossible. In addition to the two performance parameters (guess and slip), there are two learning parameters. The first is K0; the likelihood the student knows the skill when he first uses the tutor. Initial knowledge can come from a variety of places including material from a prior course or declarative instruction delivered by the tutor before the practice sessions. The second learning parameter is t; the probability a student will acquire a skill as a result of practice sessions. Every skill to be tracked based on these four parameters slip, guess, K0, and t [9].

iii. The Dirichlet Priors Approach: This approach uses a Gaussian probability distribution for parameter values [9]. To estimate Dirichlet priors, following steps are used.

1. To determine the model fitting parameters for various skill sets pertaining to a domain, use BNT-SM
2. For guess, slip, K_0, t parameters
   i. Calculate the mean (μ) and variance (σ^2) of the Parameter estimates.
   ii. Select parameters α and β to generate a Dirichlet using the Mean and Variance calculated in step1

$$\mu = \alpha/ (\alpha+\beta) \quad (5)$$

$$\sigma^2 = \alpha\beta / ((\alpha+\beta) (\alpha+\beta+1)) \quad (6)$$

   iii. Re-compute the value of guess, slip, K0 and t parameters and calculate their associated Dirichlet priors.

The extensions to the existing parameter fitting approaches are

i. Contextual Estimation of Guess and Slip (BKT-CGS) approach:

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Here the context in which parameters differ is estimated rather than having fixed values [8]. The major modifications as compared to previous approaches include:

a. Analyzing whether a given answer is a guess or slip based on log records of previous interactions.

b. The prominent actions that characterize the guess/slip parameters and which help to distinguish factors that are independent of others are done by using any machine learning approach [10].

ii. *Expectation Maximization (BKT-EM)* Approach:

There is a need to identify latent variables other than unidentified and identified parameters in order to analyze whether it is guess or a slip. A model could be designed by identifying the omitted values from the generated data or by simply guessing the existence of additional unobserved values from the data. This model fitting approach attempts to maximize the data likelihood [7]. It processes information by finding out parameters within a particular time slice, which could be termed as latent variables. It iteratively tries to maximize the likelihood by re-computing such latent variables. Rather than trying to find out global maxima, this approach tries to achieve local maxima to find the best fitting parameters [13].

iii. Brute-Force Model (BKT-BF) Approach:

Brute force is a straightforward approach for solving small–size instances of a problem. BKT-BF uses exhaustive search in problems where a particular object with specific properties has to be chosen from a set of candidate solutions. For this, all candidate solutions have to be examined extensively. Exhaustive search is not feasible if the search tree grows exponentially. For brute force model to be efficient and to reduce the computational cost, it has to make sure that the KT's parameters generate a finite search space. A BF model ensures this by defining search boundaries.When performance parameter estimates rise above 0.5, BKT-BF uses bounded guess and slip approach to avoid the model degeneracy problems.

iv. Prior per Student (BKT-PPS) Approach:

Standard BKT works on the principle that the prior knowledge of each student is same. The prior knowledge of a student can be determined from pre-tests. The BKT-PPS model modifies this concept. It divides the student knowledge into two classes-low prior and high prior. Student knowledge can be analyzed based on their first response to a practice exercise. If a student answers correctly, he is categorized into a high prior group, else low prior. Thus the prior of each group can be either set impromptu or from a trained and learned dataset. A rough classification can be done based on the assumption that a high prior should be roughly 1 minus the probability of guess and similarly the low prior should be equivalent to the probability of slip. An improved accuracy at predicting student knowledge has been improved with values 0.10 as low prior and 0.85 as high prior.

VI. OBSERVATIONS ON BKT AND ITS EXTENSIONS

Table 1 represents various approaches related to Bayesian knowledge tracing parameter fitting and discrepancies associated with them.

TABLE I. BKT PARAMETER FITTING METHODS AND THEIR DISCREPANCIES

No	BKT Parameter Fitting Methods	Issues	Selection/Optimization of parameters	
1	Baseline	Does not support parameters that change across time slices. Supports variables (observed / unobserved) having discrete values, not continuous.	Based on causal relationships between the factors that affect studentknowledge.	
2	The bounded guess and slip approach	Could be inconsistent as binding guess and slip parameters to specific range may not be accurate.	Based on the initial knowledge of the student and the probability that student acquires a skill after a practice session.	
3	Dirichlet Priors	Eventhough Dirichlet Priors are considered to be an ideal way for model fitting approaches, the issue of finding out α and β is still a concern.	Based on the student's performance data across multiple skills and calculating mean and variance.	
4	Contextual Guess and Slip	Possibility for problem of degeneracy, as parameters varies for different contexts.	Based on features of an individual action per situation or context.	
5	Expectation-Maximization (EM)	Problem of local minima and identifiability.	Based on overall student's performance data.	
6	Brute force grid search(BF)	High computational cost and problem of identifiability.	Based on the entire parameter space and by setting boundaries for exhaustive search.	
7	Prior per	Inaccuracy as	Based on the first	

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	Student (PPS)	priority is based only on the initial response.	response of the student rather than initial knowledge. Priority is decided based on correct/incorrect answer of the first question.	

VII. CONCLUSION AND FUTURE SCOPE

To help tutors to evaluate how students learn and model the tutoring system accordingly is considered as a crucial aspect in ITS. Student modelling, by which an accurate interpretation of student knowledge is predicted is done by the most popular method Bayesian Knowledge Tracing (BKT). BKT involves fitting of parameters by carefully analyzing the proficiency gained by a student on each skill. But researchers have raised concerns on validity of such parameter fitting approaches. These concerns include

a. Finding out local minima rather than global maxima, which increase computational complexity.
b. Identifiablity issue wherein multiple dissimilar dataset generating same set of conclusions.

Student modeling is considered to be a prominent building block in the field of ITS research as it helps to understand how a student has responded to a particular skill set and helps the tutor to proceed dynamically based on whether a student has acquired a skill set or not. Of the student modeling techniques mentioned above, Bayesian Knowledge Tracing is most widely accepted because of its accuracy in inferring, predicting and analyzing a student's skill set.Although many researchers have proposed many new student fitting approaches to have more accuracy in predicting and inferring a student's acquired skill set, but couldn't provide much accuracy.

Future scope includes a better parameter fitting approach that estimates the most expected and exact point at which each student has mastered a particular skill, which helps in achieving an accurate predication.

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